FACILITATOR GUIDE · CONFIDENTIAL TO THE DELIVERY TEAM

The EU Climate Machine

Run-of-show, talking points, exercise answer keys and the numbers to have at your fingertips.

Companion documents: slide deck (28 slides) · participant workbook · overview prospectus

Edition: August 2026 — refresh after every major EU policy event

How to run this day

- **Models first, slides second.** The deck exists to frame the models, not the reverse. Whenever a discussion catches fire in front of a live model, let the slides wait. Every module survives losing a third of its slides; none survives losing its exercise.
- **Predict before you run.** Each exercise forces a written prediction before touching a slider. This is deliberate: the learning is in the gap between prediction and output. Do not let teams skip to the model.
- **Honesty is the brand.** Every model is a stylised teaching instrument, and the deck says so. When a participant catches a simplification — the ETS model ignoring free allocation, the merit order having no grid — agree enthusiastically and name what the full model adds. Never defend a toy as if it were the real thing; the epistemic labels are a feature to point at, not a weakness to hide.
- **Watch the clock at the module level, not the slide level.** Hard rails: exercises start no later than 40 minutes into each module. The synthesis session is protected — never sacrifice the watchlist exercise to overrun.
- **No affiliation claims, ever.** The workshop is independently produced. If asked about ESABCC, EEA or Commission views, distinguish clearly between published positions (citable) and your own reading (yours).

Setup checklist

- Room in islands of 3–4; projector + flipchart per island if possible.
- Models preloaded and tested on the venue network **before** participants arrive; offline fallback: facilitator's laptop drives, teams call out settings.
- Print per participant: workbook. Print per team: parameter cards (workbook pp. 4, 6, 10).
- Timer visible to you; phone timers for exercise blocks.
- Half-day variant: run Modules 1 and 3 plus a 20-minute synthesis; lift the merit-order formula from Module 2 into Module 3's introduction (10 minutes, one slide).

Module 1 · The carbon pricing endgame (09:30–11:00)

Timing

Block	Minutes	What happens
Mechanism	20	Cumulative budget → Hotelling → MACC → endgame spike (slides 5–6). Check-question: "if the 2035 cap tightens, why does TODAY's price move?"
Safety valve + reality	15	Three-way minimum, sequencing stages, pipeline reality check (slides 7–8).
Exercise 1	25	Price the July 2026 package (slide 9; workbook pp. 3–5).
Debrief	20	Teams report predictions vs runs; land the five-worldviews point.
Buffer	10	Questions; spillover.

Numbers at your fingertips

Quantity	Value
Model base case price path (with CDR)	€148 (2025) · €195 (2030) · €256 (2035) · €336 (2040) · €580 (2050)
Cap-only path	€178 · €234 · €308 · €404 · €696
Published anchors it reproduces	≈€203 (2030), ≈€353 (2040) — PIK/LIMES-EU
Key defaults	cap 1,340 Mt (2025) → 0 in 2040 · bank 1,500 Mt · r 5.6%/yr · ϕmax 0.81 · CDR floor €70/t · admitted 90 Mt/yr max · haircut 10%
CDR at defaults	0 (2030) → 43 Mt (2040) → 90 Mt/yr (2050); 900 Mt cumulative
Removal costs	BECCS ≈€61–86/t · DACCS ≈€109–155/t (optimistic)
Pipeline (mid-2026)	0.2 Mt/yr operational · 1.4 Mt/yr FID · 5.8 Mt/yr announced · need 68–86 Mt/yr → ≈54×

Exercise 1 answer key

- **Mapping (step 1):** slower LRF → later net-zero year / higher cap level (price falls); 250 Mt Removals Authority → admitted volume ~25 Mt/yr from 2031, BUT on top of the cap — model it as extra admitted volume plus a small cap increase, not as in-cap substitution; MSR halving → higher effective bank; 2033 credits review → scenario branch, not a dial.
- **Expected direction (step 3):** every element is price-softening; the interesting finding is magnitude ordering — the cap/LRF change dominates, removals matter mostly after 2035, the MSR change is second-order in this model.
- **Common error to catch:** teams treating the 250 Mt as inside the cap. Art. 9c lifts the cap by 250 Mt and the Commission buys removals with the auction revenue — a Removals Authority, not an emitter-to-supplier market.
- **Discussion lever if fast:** whose worldview is each preset? Cautious sequencing = the Joule paper's own proposal; aggressive = parts of industry; no-CDR = several NGOs (~12% of consultation respondents oppose integration outright).

Questions you will get

“Isn't the EUA price just speculation?”

Speculators trade the path, not the destination: the cumulative cap pins total supply, and banking connects every year to every other. Positioning moves prices weeks; the budget moves them decades.

“Why not just cancel the surplus / tighten the MSR instead?”

Same identity, opposite sign — that IS a cap-side tightening, and the model prices it. Have a team run it live if the question lands mid-exercise.

“Is €580/t in 2050 politically survivable?”

Exactly the right question — that is why the valve exists, and why its credibility (the 54× pipeline gap) is the load-bearing uncertainty of the entire endgame. Park it visibly for Module 4.

Module 2 · Why European electricity costs what it costs (11:15–12:45)

Timing

Block	Minutes	What happens
Mechanism	20	Uniform-price auction; MC formula; draw the staircase on a flipchart (slide 11).
The comparison	20	Same machine, three fuel bills; decade of history; retail stack (slides 12–14).
Exercise 2	25	Paper merit order, then the model (slide 15; workbook pp. 6–8).
Debrief	15	The 30-second CEO answer, three volunteers deliver it.
Buffer	10	Bridge to lunch.

Numbers at your fingertips

Quantity	Value
MC formula	fuel/η + (EF/η)·CO_2 + O&M; — EF gas 0.202, coal 0.34, lignite 0.40 tCO_2/MWhth
The three bills (2025)	EU CCGT ≈€95 (67.9 fuel + 25.3 carbon + 2) · US CCGT ≈€22 · China coal ≈€36
The gap	≈€73/MWh EU–US ≈ €48 fuel + €25 carbon
Gas price-setting share	≈55–65% of EU hours (JRC 2019–22) with <20% of EU generation
TTF vs Henry Hub	structural 3–5× since 2022 (≈€34 vs €7/MWhth in 2024); LNG wedge ≈$3.3–5.6/MMBtu
Crisis pass-through	2019 gas €14→power €38 · 2022 gas €123→€235 · 2025 gas €38→€89
Retail household 2024	EU-27 28.7 ¢/kWh (49/27/24 split) · DE 39.4 ¢ · US 15.2 ¢ · CN 6.8 ¢ (industry 7.9 — inverse of EU)
Grid carbon intensity	China ≈550–600 gCO_2/kWh · EU ≈230

Exercise 2 answer key (paper stage)

With the workbook parameter card (gas €38/MWhth, coal €13, lignite €5, CO_2 €70/t):

Plant	Computation	MC €/MWh
Nuclear (78 GW)	fuel + O&M;	≈12
Hard coal (38 GW, η 0.42, EF 0.34)	13/0.42 + (0.34/0.42)·70 + 3	≈90.6
Lignite (28 GW, η 0.40, EF 0.40)	5/0.40 + (0.40/0.40)·70 + 12	≈94.5
CCGT (145 GW, η 0.56, EF 0.202)	38/0.56 + (0.202/0.56)·70 + 2	≈95.1
OCGT (45 GW, η 0.38)	38/0.38 + (0.202/0.38)·70 + 3	≈140.2

Stack at 350 GW demand with 105 GW VRE running: VRE 105 → +nuclear 78 (183) → +coal 38 (221) → +lignite 28 (249) → +CCGT 145 (394 ≥ 350). **CCGT is marginal; the price is ≈€95/MWh** — and note the coal/lignite/CCGT cluster sits within €5 of each other, which is why small fuel moves re-order the middle of the stack. August 2022 re-run: gas 235/0.56 + 25.3 + 2 ≈ **€447/MWh** for the marginal CCGT alone. US flip: 11/0.56 + 0 + 2 ≈ **€22**.

The 30-second CEO answer to listen for: “price is set hour-by-hour by the last plant needed; that plant is usually gas; our renewables lower the price when they run but cannot set it; the durable fix is less gas at the margin.”

Module 3 · Electrification and the 2040 framework (13:30–15:00)

Timing

Block	Minutes	What happens
The target	15	46% in context: benchmarks ladder, direct vs indirect, perimeter (slide 17).
The model	20	Least-cost tranches; €166 vs €55; €45 trigger; the four levers (slides 18–19).
The fallacy	15	Shrinking denominator; Liebreich's ¾ · 4/5 point (slide 20).
Exercise 3	25	Iso-target frontier: pick and defend your mix (slide 21; workbook pp. 9–11).
Debrief	15	Mixes on the flipchart; sales-vs-stock closing point.

Numbers at your fingertips

Quantity	Value
Headline trio	price-only ≈€166/t · with package ≈€55/t · gap €111/t = shadow value of demand-side policy
ETS2 trigger	€45/t (2020 prices; ≈€55 in 2025 prices); 2026 agreement doubled release volume 20→40 M allowances
Physical translation	46% = +20 pp = ≈1,620 TWh new electricity = ≈870 Mt CO_2/yr abated
Sector delivery (pp, price-only → package)	buildings 7.3→7.0 · transport 8.8→10.4 · industry 4.0→2.6
Published company	Ariadne price-only 175–350 €/t; PRIMES ETS2-2030 71–261 €/t on companion-policy strength
Heat-pump economics	beats gas boiler when electricity:gas retail ratio <~3, comfortably <~2.5; KPI ≤2.5/≤2 by 2030; only FI & SE below 2 today; NL (1.4) sells ~3× DE/PL/HU (>3)
Fallacy anchors	one boiler + one heat pump = ~23% rate at 50% device share; 46% needs ≈¾ heat pumps + ≈4/5 EV km (other sectors at zero); ~60%/65% with 25% of other fossil switched
Sensitivity	barriers ±25% → price-only 138/199, with package 45/66

Exercise 3 answer key

The workbook's precomputed inversion table (derived from the identity with COP 3.0, boiler η 0.9, ICE÷EV 3.3, heat 2,500 TWh, cars 1,700 TWh, total 8,100 TWh):

Heat-pump share	EV share needed (no other-sector help)	EV share needed (25% of other fossil switched)
50%	>100% — unreachable on this axis	≈79%
60%	≈101% — effectively unreachable	≈63%
70%	≈85%	≈48%
75% (Liebreich)	≈78%	≈40%
80%	≈70%	≈32%
90%	≈54%	≈16%

Debrief moves: (1) most teams pick a mix below the frontier — make the gap visible, then show how the 'other sectors' slider rescues it; (2) the fleet-life arithmetic: ~15-year lives mean a 2040 stock target is a 2026 sales target; (3) close on the metric debate — a target can be right while its metric misleads, and electrification may enter the target architecture as early as 2027.

Module 4 · Stress tests (15:15–16:15)

Timing

Block	Minutes	What happens
Politics priced	20	The identity; the ranked bars; free-allocation teaching point (slide 23).
Nature stress-tests back	20	Wildfire cohort logic; honest denominators; prevention arbitrage (slide 24).
Robust vs fragile	10	Two-column synthesis (slide 25).
Optional debate	10	One demand per team; defend or attack in tonnes only — no adjectives.

Numbers at your fingertips

Quantity	Value
Wishlist scenarios (cumulative to 2040)	Conservative +627 Mt (≈5.7 yrs) · Central +1,294 Mt (≈11.8 yrs; 10.4% of the 2030–50 budget) · Maximalist +2,860 Mt (≈26 yrs)
Central breakdown	LRF +671 · intl credits +280 (70% non-additionality after Cames et al. 2016) · MSR +162 · aviation +151 · free allocation +145 · CDR integrity +63 · earmark –60
Wildfire anchors	2025 record 1.08 Mha (43% in 22 Iberian fires, <2 weeks) · 5-yr mean 658 kha vs 550 kha embedded in projections
Default 2040 damage	47 Mt/yr = 44% of the planned sink improvement (212→317 Mt) = 10% of allowed 2040 net emissions = 1.0 pp of 1990
Range across presets	7 Mt (fires stop worsening) → 221 Mt (hotter fires, slower recovery) — factor ~30
Prevention arbitrage	≈77.5 tCO_2 lifecycle per avoided hectare → ≈€39/t vs €400/t engineered — ~10×; no EU instrument funds it
Member-state skew	Spain's 2030 fire loss ≈169% of its required sink improvement; 4 countries ≈75% of fire, ≈27% of sink obligation
Outer bound (no sink at all)	2050 net zero → absolute zero; €127 bn/yr engineered replacement

Facilitation notes

- Keep the wishlist module scrupulously mechanical: the model prices demands, it does not editorialise about the party making them — and say out loud that the wishlist text is a third-party summary (attribution indicative).
- The debate rule (“tonnes only, no adjectives”) is the point of the exercise: it forces the identity to do the arguing. Award the best-argued ATTACK and the best-argued DEFENCE.
- On wildfires, pre-empt the fatalism failure mode: the module's own arbitrage (€39 vs €400/t) is an agency argument, not a doom argument.

Synthesis and watchlist (16:15–17:00)

- Five numbers slide (26): present as a monitoring discipline, not a summary — each anchor pairs a number with the decision it informs.
- Watchlist exercise (20 min): each participant picks three of the five plus two organisation-specific indicators, names the source they will check quarterly, then pairs exchange and challenge (“what would make you act on this?”).

- Close with the four ideas (slide 27) — attribute each to the module where the room earned it, then the method note and thanks (slide 28).
- Collect one-line feedback cards; offer the 90-minute executive replay.

If you are running long

Sacrifice, in order: the Module 4 debate → slide 14 (retail stack; fold its one killer fact — half the bill never touches wholesale — into the debrief) → slide 19's table (teach lever 1 only). Never sacrifice: any exercise's prediction step, the pipeline reality check (slide 8), or the watchlist.